

ATTENTION!!!

- Place each section (1,2...) in one or more cells to keep the notebook organised
- Complete the codes in the `TODO` sections

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# Python Library Management System

# Install required libraries (This should already exist in colab)
!pip install requests

import pickle
import sqlite3
import requests
import os

# 1. Data Types and Variables
book_title = "Python Programming"
book_quantity = 5
book_price = 29.99
is_available = True

print(f"Book: {book_title}, Quantity: {book_quantity}, Price: ${book_price}, Available: {is_available}")

# 2. Lists
books = ["Python Basics", "Advanced Python", "Data Science with Python"]

# TODO: Add two more books to the list
books.append("Machine Learning with Python")
books.insert(2, "Web Development with Django")

print("Updated book list:", books)

# 3. Loops and range()
print("\nAvailable books:")
# TODO: Use a for loop and range() to print books with their indices
for i in range(len(books)):
    print(f"{i+1}. {books[i]}")

# 4. User Input
new_book = input("Enter a new book title: ")
books.append(new_book)
print("Updated book list:", books)

# 5. Functions and Conditional Statements
def check_availability(book_name):
    # TODO: Check if the book is in the list and return True if available, False otherwise
    if book_name in books:
        return True
    else:
        return False

# Test the function
print(f"Is 'Python Basics' available? {check_availability('Python Basics')}")

# 6. File I/O
def save_books_to_file(filename):
    with open(filename, 'w') as f:
        for book in books:
            f.write(f"{book}\n")

# TODO: Implement a function to read books from a file
def read_books_from_file(filename):
    try:
        with open(filename, 'r') as f:
            return [line.strip() for line in f.readlines()]
    except FileNotFoundError:
        print(f"File {filename} not found.")
        return []

# Test file I/O
save_books_to_file("books.txt")
!cat books.txt # Display file contents in Colab

# 7. Pickle for serialization
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def save_books_pickle(filename):
    with open(filename, 'wb') as f:
        pickle.dump(books, f)

# TODO: Implement a function to load books using pickle
def load_books_pickle(filename):
    try:
        with open(filename, 'rb') as f:
            return pickle.load(f)
    except FileNotFoundError:
        print(f"File {filename} not found.")
        return []

# Test pickle functions
save_books_pickle("books.pkl")
print("Books loaded from pickle:", load_books_pickle("books.pkl"))

# 8. SQLite Database Interaction
def create_books_table():
    conn = sqlite3.connect('library.db')
    cursor = conn.cursor()
    cursor.execute('CREATE TABLE IF NOT EXISTS books
                    (id INTEGER PRIMARY KEY, title TEXT, quantity INTEGER)')
    conn.commit()
    conn.close()

# TODO: Implement a function to add a book to the SQLite database
def add_book_to_db(title, quantity):
    conn = sqlite3.connect('library.db')
    cursor = conn.cursor()
    cursor.execute("INSERT INTO books (title, quantity) VALUES (?, ?)", (title, quantity))
    conn.commit()
    conn.close()
    print(f"Added '{title}' to database with quantity {quantity}")

# Test database functions
create_books_table()
add_book_to_db("Python for Beginners", 10)

# 9. Web API Interaction (GitHub API)
def get_python_repos():
    url = "https://api.github.com/search/repositories"
    params = {"q": "language:python", "sort": "stars", "order": "desc"}
    response = requests.get(url, params=params)
    data = response.json()
    return [repo['name'] for repo in data['items'][:5]]

# Test API function
print("Top 5 Python repositories on GitHub:", get_python_repos())

# ===== NEW FEATURES: Removing books, Updating books, CLI Interface =====

# 10. Remove books from the library
def remove_book(book_name):
    """Remove a book from the library by name"""
    if book_name in books:
        books.remove(book_name)
        print(f"Removed '{book_name}' from the library")
        return True
    else:
        print(f"Book '{book_name}' not found in the library")
        return False

# Remove from database as well
def remove_book_from_db(book_title):
    conn = sqlite3.connect('library.db')
    cursor = conn.cursor()
    cursor.execute("DELETE FROM books WHERE title = ?", (book_title,))
    conn.commit()
    affected_rows = cursor.rowcount
    conn.close()
    if affected_rows > 0:
        print(f"Removed '{book_title}' from database")
        return True
    else:
        print(f"Book '{book_title}' not found in database")
        return False

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# 11. Update book information
def update_book(old_title, new_title=None, new_quantity=None):
    """Update book title and/or quantity in the list and database"""
    # Update in the list
    if old_title in books:
        idx = books.index(old_title)
        if new_title and new_title != old_title:
            books[idx] = new_title
            print(f"Updated title from '{old_title}' to '{new_title}'")
        else:
            print(f"Book '{old_title}' found in library")
    else:
        print(f"Book '{old_title}' not found in library")
        return False

    # Update in database
    conn = sqlite3.connect('library.db')
    cursor = conn.cursor()

    if new_title and new_quantity:
        cursor.execute("UPDATE books SET title = ?, quantity = ? WHERE title = ?",
                       (new_title, new_quantity, old_title))
    elif new_title:
        cursor.execute("UPDATE books SET title = ? WHERE title = ?",
                       (new_title, old_title))
    elif new_quantity:
        cursor.execute("UPDATE books SET quantity = ? WHERE title = ?",
                       (new_quantity, old_title))
    else:
        print("No updates provided")
        conn.close()
        return False

    conn.commit()
    affected_rows = cursor.rowcount
    conn.close()

    if affected_rows > 0:
        print("Database updated successfully")
    else:
        print("Book not found in database")

    return True

# 12. Simple command-line interface for user interaction
def display_menu():
    """Display the main menu"""
    print("\n" + "="*50)
    print("LIBRARY MANAGEMENT SYSTEM")
    print("="*50)
    print("1. View all books")
    print("2. Add a book")
    print("3. Remove a book")
    print("4. Update book information")
    print("5. Check book availability")
    print("6. Save books to file")
    print("7. Load books from file")
    print("8. View database contents")
    print("9. Exit")
    print("="*50)

def view_books():
    """Display all books in the library"""
    if not books:
        print("\nThe library is empty.")
    else:
        print("\nBooks in the library:")
        for i, book in enumerate(books, 1):
            print(f"{i}. {book}")

def view_database():
    """Display all books in the SQLite database"""
    conn = sqlite3.connect('library.db')
    cursor = conn.cursor()
    cursor.execute("SELECT * FROM books")
    rows = cursor.fetchall()
    print(rows)

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conn.close()

if not rows:
    print("\nDatabase is empty.")
else:
    print("\nBooks in database:")
    print("ID | Title | Quantity")
    print("-" * 30)
    for row in rows:
        print(f"{row[0]} | {row[1]} | {row[2]}")

def main():
    """Main CLI interface"""
    print("Welcome to the Library Management System!")

    # Load existing database if available
    create_books_table()

    while True:
        display_menu()
        choice = input("\nEnter your choice (1-9): ")

        if choice == '1':
            view_books()

        elif choice == '2':
            title = input("Enter book title: ")
            if title in books:
                print(f"'{title}' already exists in the library.")
            else:
                books.append(title)
                quantity = int(input("Enter quantity: "))
                add_book_to_db(title, quantity)
                print(f"Added '{title}' to the library.")

        elif choice == '3':
            title = input("Enter book title to remove: ")
            remove_book(title)
            remove_book_from_db(title)

        elif choice == '4':
            old_title = input("Enter current book title: ")
            new_title = input("Enter new title (press Enter to keep same): ")
            quantity_str = input("Enter new quantity (press Enter to keep same): ")

            if not new_title:
                new_title = None
            if quantity_str:
                new_quantity = int(quantity_str)
            else:
                new_quantity = None

            update_book(old_title, new_title, new_quantity)

        elif choice == '5':
            title = input("Enter book title to check: ")
            if check_availability(title):
                print(f"'{title}' IS available in the library.")
            else:
                print(f"'{title}' is NOT available in the library.")

        elif choice == '6':
            filename = input("Enter filename to save to (default: books.txt): ") or "books.txt"
            save_books_to_file(filename)
            print(f"Books saved to {filename}")

        elif choice == '7':
            filename = input("Enter filename to load from (default: books.txt): ") or "books.txt"
            loaded_books = read_books_from_file(filename)
            if loaded_books:
                books.extend(loaded_books)
                print(f"Loaded {len(loaded_books)} books from {filename}")
                view_books()

        elif choice == '8':
            view_database()

        elif choice == '9':

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        print("Thank you for using the Library Management System!")
        break

    else:
        print("Invalid choice. Please enter a number between 1 and 9.")

# Run the main interface (commented out for safety)
# Uncomment the line below to run the CLI
# if __name__ == "__main__":
#     main()

print("\nLibrary Management System operations completed.")

# Demo of new features
print("\n" + "="*50)
print("DEMO OF NEW FEATURES:")
print("="*50)

# Demo: Remove a book
print("\n1. Removing a book:")
remove_book("Machine Learning with Python")

# Demo: Update a book
print("\n2. Updating book information:")
update_book("Data Science with Python", "Python Data Science", 7)

# Demo: Show updated list
print("\n3. Updated book list:")
view_books()

# Demo: Show database
print("\n4. Database contents:")
view_database()

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Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests) (2026.7.22)
Book: Python Programming, Quantity: 5, Price: \$29.99, Available: True
Updated book list: ['Python Basics', 'Advanced Python', 'Web Development with Django', 'Data Science with Python', 'Machine Learning with Python']

Available books:
1. Python Basics
2. Advanced Python
3. Web Development with Django
4. Data Science with Python
5. Machine Learning with Python
Enter a new book title: Python Programming
Updated book list: ['Python Basics', 'Advanced Python', 'Web Development with Django', 'Data Science with Python', 'Machine Learning with Python', 'Python Programming']
Is 'Python Basics' available? True
Python Basics
Advanced Python
Web Development with Django
Data Science with Python
Machine Learning with Python
Python Programming
Books loaded from pickle: ['Python Basics', 'Advanced Python', 'Web Development with Django', 'Data Science with Python', 'Machine Learning with Python', 'Python Programming']
Added 'Python for Beginners' to database with quantity 10
Top 5 Python repositories on GitHub: ['public-apis', 'free-programming-books', 'system-design-primer', 'awesome-python', 'project-awesome']

Library Management System operations completed.

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DEMO OF NEW FEATURES:

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1. Removing a book:
Removed 'Machine Learning with Python' from the library

2. Updating book information:
Updated title from 'Data Science with Python' to 'Python Data Science'
Book not found in database

3. Updated book list:

Books in the library:
1. Python Basics
2. Advanced Python
3. Web Development with Django
4. Python Data Science

5. Python Programming

4. Database contents:

Books in database:

ID	Title	Quantity
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1	Python for Beginners	10
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